AXOLABS METHOD VALIDATION REPORT FOR THE ANALYSIS OF 2439-EXNA IN DOG TISSUE	
Axolabs Project Number	AN406.0-24
Project Title	Validation of an AEX-HPLC-based method for the analysis of 2439-exNA in dog tissue
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Start Date of Bioanalytical Test Phase	14-Nov-2024
End Date of Bioanalytical Test Phase	22-Nov-2024

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Name and Function	Date and Signature
Alissa Coffey **SPONSOR'S REPRESENTATIVE**	Signed by: [redacted] Signer Name: Alissa Coffey Signing Reason: I approve this document Signing Time: 21-Jan-2025 | 15:21 CET 3E846AC45A5F4A42B5DCE31B3306D2F1

COMPLIANCE STATEMENT

The method validation was conducted following sound scientific principles and applying the acceptance criteria as given in *ICH guideline M10 on bioanalytical method validation and study sample analysis* (EMA/CHMP/ICH/172948/2019; European Medicines Agency; 21 January 2023). No compliance is claimed.

Study Director: Nicole Rottmann

Signiert von:

[redacted]

Name des Unterzeichners: Nicole Rottmann
Signiergrund: Ich habe dieses Dokument verfasst
Signierzeit: 20-Jan-2025 | 14:24 MEZ

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Name | Date and Signature

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LIST OF ABBREVIATIONS

AB0X	Analytical Batch, AN406.0-24_01_AB0X
ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
BQL	Below Quantification Limit (Below the LLOQ)
CHMP	Committee for Medicinal Products for Human Use
CoA	Certificate of Analysis
CSF	Cerebrospinal Fluid
CV	Coefficient of Variation
EDTA	Ethylenediaminetetraacetic Acid
EMA	European Medicines Agency
GCP	Good Clinical Practice
GLP	Good Laboratory Practice
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
LLOQ	Lower Limit of Quantification
LTS	Long-Term Stability
MDR	Method Development Report
PBS	Phosphate Buffered Saline
PNA	Peptide Nucleic Acid
PrP	Prion Protein
QC	Quality Control
SDS	Sodium Dodecyl Sulfate
siRNA	small interfering Ribonucleic Acid
SOP	Standard Operating Procedure
SST	System Suitability Test(ing)
STM	Standard Test Method
TCEP	Tris(2-carboxyethyl) phosphine hydrochloride solution
ULOQ	Upper Limit of Quantification
UV	Ultraviolet

1 SUMMARY

This report summarizes the results of the bioanalytical method validation for the analysis of 2439-exNA in dog liver, kidney and brain (Axolabs Project: AN406.0-24).

Brain is the tissue of main concern and will therefore be tested more thoroughly than the secondary matrices liver and kidney, in which only selectivity, accuracy, precision and matrix effect were assessed.

The assay is based on the specific hybridization of the antisense strands of 2439-exNA, a dimer composed of two identical siRNAs, which are linked at their sense strands with a tetraethylene glycol linker, with a complementary peptide nucleic acid (PNA) strand in a 1:1 stoichiometry relative to each antisense strand. The PNA is labeled with an Atto425 fluorescence dye and yields a specific signal in the subsequent analysis by Peptide Nucleic Acid Anion-Exchange High-Performance Liquid Chromatography (PNA-AEX-HPLC) coupled to a fluorescence detector.

The method was developed without the need for any analyte extraction from dog tissue. Therefore, recovery of 100 % is maintained over the process (no internal standard is needed).

The bioanalytical method was previously developed at Axolabs GmbH and is described in the Axolabs Method Development Report (MDR) *MDR_2439exNA_DogTissue* (edition 01, effective date: 24-Oct-2024) that was used as the basis for this method validation (**Supplemental File 1**). During the method development phase, the suitability of the method for dog tissue was shown.

The method validation was carried out with the same conditions and parameters described in the MDR, except for the preparation of reference standard. In the method development a lyophilized reference standard was used, which was weighed and dissolved. In the validation, an already prepared reference standard solution provided by the Sponsor was used.

The final conditions of the validated method are described in the Standard Test Method (STM) *STM_Broad_2439-exNA_DogTissue* (edition 02, effective date: 03-Dec-2024; **Supplemental File 2**).

Additional bioanalytical method validations for 2439-exNA conducted at Axolabs GmbH are listed in Table 1.

Table 1: Summary Information of Additional Method Validations for 2439-exNA

Axolabs Project	Type of Assay	Details of Validation	Date of Final Report
AN402.0-24	PNA-HPLC-Assay with Fluorescence Detection	other matrix (dog plasma)	04-Dec-2024
AN403.0-24	PNA-HPLC-Assay with Fluorescence Detection	other species and matrix (rat plasma)	03-Dec-2024
AN404.0-24	PNA-HPLC-Assay with Fluorescence Detection	other matrix (dog CSF)	18-Dec-2024
AN405.0-24	PNA-HPLC-Assay with Fluorescence Detection	other species and matrix (rat CSF)	18-Dec-2024

The work was done according to the Method Validation Plan signed on 13-Nov-2024 by the Sponsor's Representative.

The results of the method validation are summarized in Table 2.

Table 2: Overall Summary Table

Items	Results	Section/Table	Comments
Analyte	2439-exNA	Section 3	n.a.
Reference Standard	2439-exNA	Section 4	n.a.
Methodology	PNA-AEX-HPLC-Assay with Fluorescence Detection	Section 8.1	n.a.
Biological Matrix	Dog tissue	Section 5 Table 3	n.a.
Calibration Curve Range	20.0 ng/g – 10,000.0 ng/g	Section 9.2 Table 9	n.a.
Selectivity	5 times higher response (peak area) in the spiked QC-LLOQ samples compared to the signal in the blank samples (for each tissue type)	Section 9.3 Table 11, Table 12 and Table 13	n.a.
Matrix Effect	No matrix effect observed for 6 individually tested matrix lots (for each tissue type)	Section 9.4 Table 14, Table 15 and Table 16	n.a.
Carry-Over	<20 % of the response (peak area) of the LLOQ calibration standard	Section 9.5 Table 17	n.a.
Between-Run Accuracy (for brain tissue)	QC-LLOQ (20.0 ng/g): 104.3 % QC-Low (60.0 ng/g): 93.7 % QC-Mid (4,500.0 ng/g): 99.4 % QC-High (9,000.0 ng/g): 99.6 %	Section 9.7.2 Table 18	n.a.
Between-Run Accuracy (for kidney tissue)	QC-LLOQ (20.0 ng/g): 99.7 % QC-Low (60.0 ng/g): 93.1 % QC-Mid (4,500.0 ng/g): 99.2 % QC-High (9,000.0 ng/g): 98.9 %	Section 9.7.2 Table 19	n.a.
Between-Run Accuracy (for liver tissue)	QC-LLOQ (20.0 ng/g): 101.2 % QC-Low (60.0 ng/g): 94.7 % QC-Mid (4,500.0 ng/g): 99.3 % QC-High (9,000.0 ng/g): 99.2 %	Section 9.7.2 Table 20	n.a.
Between-Run Precision (for brain tissue)	QC-LLOQ (20.0 ng/g): 8.6 % QC-Low (60.0 ng/g): 4.1 % QC-Mid (4,500.0 ng/g): 2.2 % QC-High (9,000.0 ng/g): 2.6 %	Section 9.8.2 Table 18	n.a.
Between-Run Precision (for kidney tissue)	QC-LLOQ (20.0 ng/g): 8.8 % QC-Low (60.0 ng/g): 3.7 % QC-Mid (4,500.0 ng/g): 1.8 % QC-High (9,000.0 ng/g): 2.4 %	Section 9.8.2 Table 19	n.a.

Items	Results	Section/Table	Comments
Between-Run Precision (for liver tissue)	QC-LLOQ (20.0 ng/g): 8.1 % QC-Low (60.0 ng/g): 2.9 % QC-Mid (4,500.0 ng/g): 1.8 % QC-High (9,000.0 ng/g): 2.2 %	Section 9.8.2 Table 20	n.a.
Dilution Integrity	Dilution QC: 5,000,000.0 ng/g (Dilution factor: 1,000) Accuracy: 86.2 % Precision: 12.1 %	Section 9.9 Table 21	n.a.
Long-Term Stability	Ongoing (T=0) at -80 °C ± 20 °C (in homogenate and lysate)	Section 9.10.1 Table 22 and Table 23	n.a.
Freeze-Thaw Stability	Demonstrated for 5 cycles at -80 °C ± 20 °C	Section 9.10.2 Table 24	n.a.
Bench-Top (Robustness) and Short-Term Stability	Demonstrated for 4 hours at room temperature	Section 9.10.4 Table 25	n.a.
Processed Sample Stability	Demonstrated for 119 hours at 20 °C ± 3 °C	Section 9.10.5 Table 26	n.a.
Short-Term Stability of Lowest Concentration of the Working Solution	Demonstrated for 4 hours at room temperature	Section 9.10.6.1 Table 27	n.a.
Long-Term Stability of Stock Solution	Was not performed as data were provided by the Sponsor		

2 COMPLIANCE

The method validation was conducted following sound scientific principles and applying the acceptance criteria as given in *ICH guideline M10 on bioanalytical method validation and study sample analysis* (EMA/CHMP/ICH/172948/2019; European Medicines Agency; 21 January 2023). No compliance is claimed.

3 Analyte

The analyte 2439-exNA is a prion protein (PrP)-lowering divalent siRNA (**Fehler! Verweisquelle konnte nicht gefunden werden.**). The sense strands of the two identical siRNAs are linked to each other via a tetraethylene glycol linker.

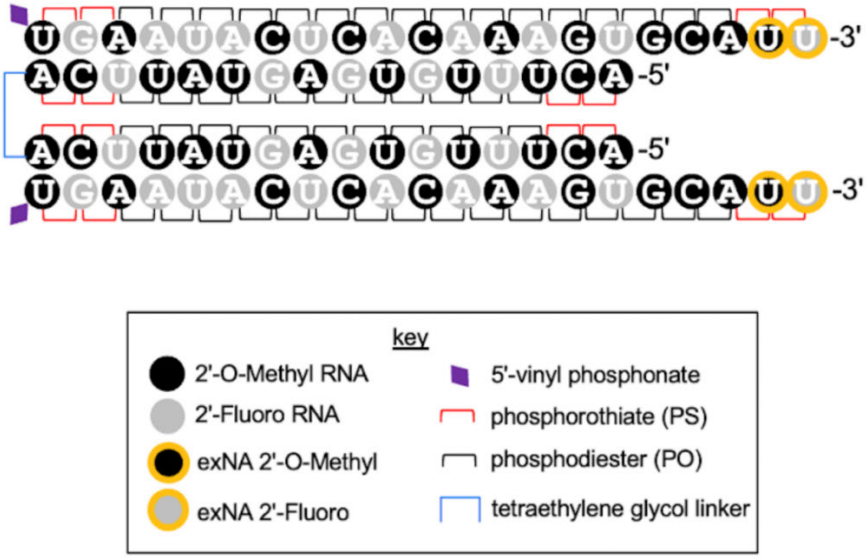


Figure 1: Structure of the PrP-lowering divalent siRNA 2439-exNA

4 Reference Standard

Identification:	PrP-lowering divalent siRNA 2439-exNA (shortened to 2439-exNA in this document)
Structure:	Refer to Section 3
Physical Description:	Slightly yellow solution
Purity:	96.6 % (determined by non-denaturing HPLC)
Total Oligo Concentration:	9.9 mg/mL (determined by UV spectrophotometry)
Full-Length Concentration:	9.5634 mg/mL (corrected for purity stated on the CoA from 19-Aug-2024)
Supplier:	Argonaut Manufacturing Services (Carlsbad, CA, USA)
Lot No.:	240621005
Manufacturing Date:	24-Jun-2024
Retest Date:	24-Jun-2026

With the exception of a document-based inspection and control of the corresponding Certificate of Analysis (CoA), no further testing on identity and purity of the reference standard was conducted upon receipt.

The reference standard was provided by Argonaut Manufacturing Services as a liquid formulation and could be used directly to prepare the working solutions (using 0.1 % SDS as diluent) for the calibration standards and the Quality Control (QC) samples.

Separate aliquots of the reference standard were used for the calibration standards and the Quality Control (QC) samples. For this purpose, the reference standard was previously aliquoted into 100 µL aliquots (Eppendorf Protein LoBind tubes). After aliquoting, the reference standard was renamed to 2439-exNA_240621005_Stock1_04-Sep-2024 in accordance with internal nomenclature.

The concentrations of calibration standards and QC samples refer to the full-length concentration of 2439-exNA.

5 Blank Matrices

Blank matrices used in this method validation are listed in Table 3.

Table 3: Blank Matrices

Blank Matrix	Sex	Receipt Date	Supplier	Lot No.	Expiry Date[1)] (Lysates)	Analysis Batch
Dog brain	Male	10-Oct-2024	AmplifyBio, West Jefferson, Ohio, USA	101	12-Nov-2025	AB01 – AB03
	Male	10-Oct-2024		102	12-Nov-2025	AB01
	Male	10-Oct-2024		103	12-Nov-2025	AB01, AB02
	Female	10-Oct-2024		151	12-Nov-2025	AB01, AB02
	Female	10-Oct-2024		152	12-Nov-2025	AB01, AB03
	Female	10-Oct-2024		153	12-Nov-2025	AB01, AB03
	n.a.	19-Sep-2024		Broad Prep1	12-Nov-2025	AB01 – AB03
	n.a.	19-Sep-2024		Broad Prep2	12-Nov-2025	AB01 – AB03
	n.a.	19-Sep-2024		Broad Prep3	12-Nov-2025	AB01 – AB03
	n.a.	19-Sep-2024		Broad Prep5	12-Nov-2025	AB01 – AB03
	n.a.	19-Sep-2024		Broad Prep6	12-Nov-2025	AB01 – AB03
Dog liver	Male	10-Oct-2024		101	12-Nov-2025	AB01
	Male	10-Oct-2024		102	12-Nov-2025	AB01
	Male	10-Oct-2024		103	12-Nov-2025	AB01 – AB02
	Female	10-Oct-2024		151	12-Nov-2025	AB01 – AB02
	Female	10-Oct-2024		152	12-Nov-2025	AB01, AB03
	Female	10-Oct-2024		153	12-Nov-2025	AB01 – AB03
Dog kidney	Male	10-Oct-2024		101	12-Nov-2025	AB01
	Male	10-Oct-2024		102	12-Nov-2025	AB01

Blank Matrix	Sex	Receipt Date	Supplier	Lot No.	Expiry Date[1)] (Lysates)	Analysis Batch
	Male	10-Oct-2024		103	12-Nov-2025	AB01 – AB02
	Male	10-Oct-2024		104	12-Nov-2025	AB01
	Male	10-Oct-2024		105	12-Nov-2025	AB01
	Male	10-Oct-2024		106	12-Nov-2025	AB01
	Female	10-Oct-2024		151	12-Nov-2025	AB02
	Female	10-Oct-2024		152	12-Nov-2025	AB03
	Female	10-Oct-2024		153	12-Nov-2025	AB03

[1)] In accordance with KU-RGX-SOP-004, the expiry date for solid matrices is 5 years from the date of receipt.

6 PNA Probe

PNA Probe: PNA-BroadInst-01 from HLB Panagene Co., Ltd. (Daejeon, South Korea):

- N-terminus-Atto425 – OO – cdc ttt gtg dgt d – C-terminus, complementary to the antisense strands of 2439-exNA (OO: polyethylene glycol linker; d: 2,6-diaminopurine base modification)
  - Bulk powder was dissolved in 10 % acetonitrile (ACN) and 0.01 % Tween 20
  - Stock solution was diluted with 10 % ACN and 0.01 % Tween 20 to 2 µM (working solution)
- Lot No.: 241567
- Expiry Date: 02-Oct-2027

7 Reference Standard, PNA Probe, and Blank Matrices – Inventory and Disposition

Records of the receipt, distribution and storage of the reference standard, PNA probe, and blank matrices were maintained.

Unused reference standard will be retained for use in subsequent studies for the Sponsor.

Unused PNA probe and blank matrices will be retained for use in subsequent studies.

8 Method

8.1 Method Summary

This report describes the validation of an AEX-HPLC method with fluorescence detection that allows the sensitive and specific detection of the antisense strands of 2439-exNA in dog tissue samples.

The assay is based on the specific hybridization of a 13-mer PNA probe conjugated at the N-terminus with an Atto425 dye that is complementary to the antisense strands of the divalent siRNA 2439-exNA.

In the first step of the assay, dog tissue lysates are prepared by Proteinase K treatment. Then, the antisense strands of 2439-exNA will be hybridized with the PNA and subsequently analyzed by AEX-HPLC coupled to a fluorescence detector.

Quantification was performed based on an external calibration curve generated from a standard dilution series in dog brain lysates. Linear calibration curves (weighted 1/X) were calculated from 20.0 ng/g to 10,000.0 ng/g.

The detailed procedure is described in the STM, which is included as a supplemental file to this report.

8.2 Applicable Documents

- Axolabs Method Validation Plan for the analysis of 2439-exNA in dog tissue (Axolabs Project: AN406.0-24)
- Method Development Report (MDR) *MDR_2439exNA_DogTissue* (edition: 01, effective date: 24-Oct-2024)
- Standard Test Method (STM) *STM_Broad_2439-exNA_DogTissue* (edition 02, effective date: 03-Dec-2024)
- KU-RGL-SOP-002, Revision 03 (formerly: REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*)
- KU-RGL-SOP-009, Revision 04 (formerly SOP REG-109) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*)
    - KU-RGL-INS-005, Revision 05 (formerly REG-109-I01) – PNA-HPLC-Assay SST
- KU-RGL-SOP-010, Revision 01 (formerly: REG-110) – Herstellung von Standardlösungen mit einer Präzisionswaage (*Preparation of Standard Solutions using a Precision Balance*)

8.3 Computerized Systems

Critical computerized systems used in this method validation are listed in Table 4.

Table 4: Computerized Systems

System Name	Revison / Version	System Description
Thermo Fisher Ultimate 3000 HPLC System with a Shimadzu RF-20Axs Fluorescence Detector used with Chromeleon HPLC Software	7.2 SR5	HPLC Data collection and export to Excel

System Name	Revison / Version	System Description
Microsoft Excel Spreadsheet: KU-RGL-ESS-002 (formerly REG-102-S02) – Method Validation Data Report	03	Calculates results of calibration standard and QC sample accuracy and precision
Microsoft Excel Spreadsheet: KU-RGL-ESS-003 (formerly: REG-102-S03) – Method Validation Protocol PNA Assay - Solid Matrices	01	Used as instruction and as the protocol for the assay execution for solid matrices: Calculates all sample preparation steps and the HPLC sequence for an analysis batch
Microsoft Excel Spreadsheet: KU-RGL-ESS-014 (formerly REG-109-S04 – Preparation of HPLC Buffers & Stock Solutions	02	Calculates all buffer preparation steps, including the preparation of the required stock solutions

8.4 HPLC Systems and Parameters

The HPLC systems used for the AEX-HPLC analyses are listed in Table 5.

Table 5: HPLC Systems

HPLC System	Module	Supplier	Model	Serial No.
HPLC-GLP-9 (AB02 and AB03)	Degasser/Solvent Rack	Thermo Fisher Scientific	Ultimate SR-3000	7316320
	Pump		Ultimate LPG-3400SD	8158985
	Autosampler		Ultimate WPS-3000TSL	8159066
	Column Oven		Ultimate TCC-3000RS	6022996
	Fluorescence Detector	Shimadzu	RF-20AXS	L20505673998
	Communication Unit		CBM-20A	L20235670992
HPLC System	**Module**	**Supplier**	**Model**	**Serial No.**
HPLC-GLP-11 (AB01)	Degasser/Solvent Rack	Thermo Fisher Scientific	Ultimate SR-3000	7327605
	Pump		Ultimate LPG-3400SD	8170487
	Autosampler		Ultimate WPS-3000TSL	8170587
	Column Oven		Ultimate TCC-3000RS	6026712
	Fluorescence Detector	Shimadzu	RF-20AXS	L20505774076
	Communication Unit		CBM-20A	L20235771316

- Eluent Buffer A: 25 mM Trizma hydrochloride solution (pH 8), 1 mM EDTA, 30 % ACN, 3 M urea
- Eluent Buffer B: 25 mM Trizma hydrochloride solution (pH 8), 1 mM EDTA, 30 % ACN, 3 M urea, 800 mM NaClO4
- Eluent Buffer C: 10 % ACN, 4 M NaClO4, 2 M urea, 0.5 x triphosphate buffer, adjusted to pH 2.5 with phosphoric acid
- Eluent Buffer D: 50 % ACN
- Flow Rate: 1 mL/min
- Detection: Excitation @ 436 nm; Emission @ 484 nm; Gain: Medium 16
- Injection of 50 µL of the hybridized sample (corresponds to 0.1 mg matrix on column)
- Column Temperature: 45 °C
- Autosampler Temperature: 20 °C
- HPLC Program: Broad_dogtissue

Table 6: HPLC Gradient Table

Time [min]	% Eluent A	% Eluent B	% Eluent C	% Eluent D	Curve
0.00	86	14	0	0	linear
1.00	86	14	0	0	linear
6.00	76	24	0	0	linear
6.10	0	0	0	100	linear
9.10	0	0	0	100	linear
9.20	86	14	0	0	linear
11.20	86	14	0	0	linear
11.30	0	0	100	0	linear
12.30	0	0	100	0	linear
12.40	86	14	0	0	linear
12.90	86	14	0	0	linear
13.00	0	0	100	0	linear
14.00	0	0	100	0	linear
14.10	86	14	0	0	linear
14.60	86	14	0	0	linear
14.70	0	0	100	0	linear
15.70	0	0	100	0	linear
15.80	86	14	0	0	linear
20.00	86	14	0	0	linear

8.5 HPLC Columns

Thermo Fisher Scientific DNAPac PA200 (4 x 250 mm) Column; Serial Number 010628; Lot Number 02306174 was used for the analytical batch AB01.

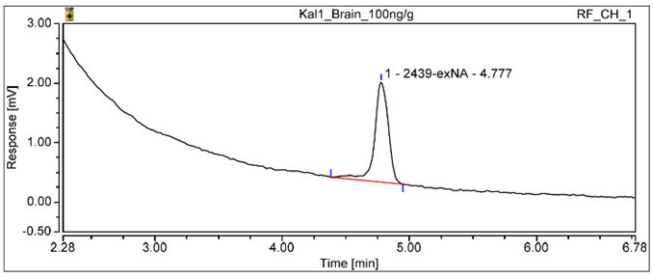
Thermo Fisher Scientific DNAPac PA200 (4 x 250 mm) Column; Serial Number 011142; Lot Number 02321141 was used for the analytical batch AB02 and AB03.

8.6 Data Evaluation

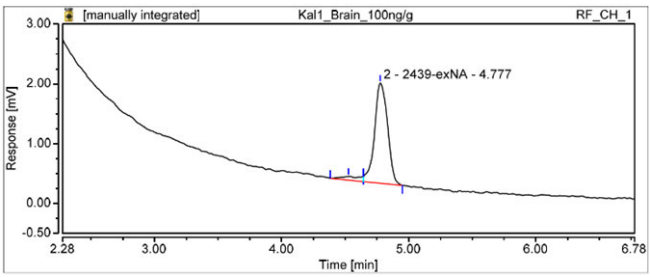
Data acquisition and integration were performed using the software package Chromeleon (Version 7.2) from Thermo Fisher Scientific.

For every injection, the peaks were integrated automatically by the HPLC chromatography software but are adjusted manually if the baseline and peak shoulders are not detected correctly by the HPLC software. When necessary, manual integration was performed by vertical peak splits at minima between two not fully resolved peaks. General rules for manual integration are documented in KU-RGL-SOP-009 (formerly SOP REG-109).

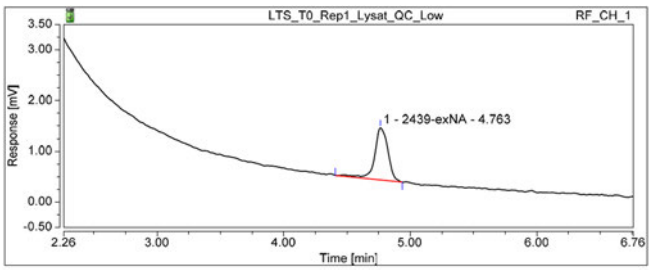
A report format of this file is generated containing values of retention time, peak area, and concentration of 2439-exNA in dog tissue and was saved in PDF format.



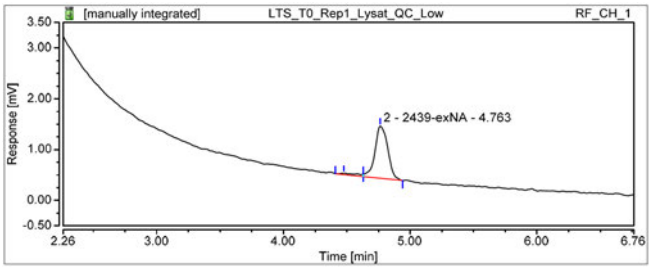
HPLC chromatogram of Cal-level 3 – 100.0 ng/g, sample no. 7 in AB01 (original)



HPLC chromatogram of Cal-level 3 – 100.0 ng/g, sample no. 7 in AB01 **(re-integrated)**



HPLC chromatogram of LTS sample (T= 0), sample no. 21 in AB01 **(original)**



HPLC chromatogram of LTS sample (T= 0), sample no. 21 in AB01 **(re-integrated)**

Figure 2: Representative HPLC chromatograms (original and re-integrated)

8.7 Statistical Analysis

Mean values, standard deviations (SD), and coefficients of variation (CV) were calculated using validated Microsoft Excel spreadsheets.

9 Results and Discussion

An overview of all performed analytical batches during the method validation is listed in Table 7.

9.1 System Suitability Test (SST)

The reproducibility of the HPLC system was determined by conducting an SST according to KU-RGL-INS-005 (formerly REG-109-I01) before measuring an analysis batch. Acceptance criteria are described in KU-RGL-INS-005 (formerly REG-109-I01), and the results of the SST run are attached to the documentation of each analysis batch.

A successful SST was completed before each reported analysis batch.

9.2 Calibration Curve

Calibration standards were generated by spiking known concentrations of 2439-exNA into blank dog brain lysates. The linear range was determined from 20.0 ng/g to 10,000.0 ng/g. The calibration levels used were 20.0 ng/g, 100.0 ng/g, 500.0 ng/g, 1,000.0 ng/g, 5,000.0 ng/g, 10,000.0 ng/g and one blank sample (dog tissue lysate processed in the absence of 2439-exNA).

The calibration curves were prepared in duplicate for each analysis batch. Blank samples and calibration standards not meeting the acceptance criteria were not used to calculate the calibration curves.

In total, 6 calibration curves were analyzed in 3 different analysis batches.

9.2.1 Concentration-Response

The concentration-response relation was calculated by linear regression of the mean peak responses of the 6 calibration standards using 1/X weighting.

The calibration curve parameters (slope, y-axis offset, and coefficient of determination) from reported analysis batches are listed in Table 8.

9.2.2 Calibration Curve Accuracy

Acceptance Criteria:

The back-calculated concentrations of the calibration standards should be within ± 15 % of the nominal concentrations (± 20 % for the LLOQ calibration standard).

At least 75 % of the calibration standards, with a minimum of 6 calibration standard levels, and at least 50 % of the calibration standards per concentration level must fulfill this criterion.

Results:

- 100 % of the back-calculated concentrations for all calibration standards were within ± 15 % (± 20 % for the LLOQ calibration standard) of the nominal concentrations.

Data are listed in Table 9.

Based on these results, the method can be used to determine concentrations of 2439-exNA in dog tissue over the concentration range of 20.0 ng/g to 10,000.0 ng/g.

9.2.3 Sensitivity/Lower Limit of Quantification (LLOQ)

The sensitivity of the assay is defined as the lowest 2439-exNA concentration that can be quantified with acceptable accuracy and precision – i.e., the LLOQ.

To confirm the LLOQ, 6 individually prepared samples with a nominal concentration at the LLOQ were analyzed (LLOQ calibration standards).

Acceptance Criteria:

The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same calibration curve.

The mean back-calculated concentration of the 6 LLOQ calibration standards should be within ± 20 % of the nominal concentration and the coefficient of variation (CV) should not exceed 20 %.

Results:

- No signal at the retention time of 2439-exNA was detected in the blank samples of the calibration curves and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same calibration curve.
- The mean back-calculated concentration of the 6 LLOQ calibration standards was 102.2 % of the nominal concentration with a CV of 3.6 %.

Data are listed in Table 9 and Table 10.

The LLOQ for 2439-exNA was confirmed at 20.0 ng/g in dog tissue.

9.2.4 Upper Limit of Quantification (ULOQ)

The highest calibration standard in the calibration curve defines the ULOQ.

Six (6) individually prepared samples with a nominal concentration at the ULOQ were analyzed (ULOQ calibration standards).

Acceptance Criteria:

The mean back-calculated concentration of the 6 ULOQ calibration standards should be within ± 15 % of the nominal concentration and the CV should not exceed 15 %.

Results:

- The mean back-calculated concentration of the 6 ULOQ calibration standards was 100.0 % of the nominal concentration with a CV of 0.3 %.

Data are listed in Table 9.

The ULOQ for 2439-exNA was confirmed at 10,000.0 ng/g in dog tissue.

9.3 Selectivity

The selectivity of the method was evaluated by comparison of unspiked lysate samples of 6 different blank dog tissue lots with the same dog tissue lysate samples spiked with 2439-exNA at the concentration of the LLOQ (QC-LLOQ).

Acceptance Criteria:

The analyte response (peak area) in each QC-LLOQ sample should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same matrix lot.

The back-calculated concentration of each QC-LLOQ sample should be within ± 20 % of the nominal concentration.

Results for Brain Tissue:

- In 1 of 6 selectivity samples (Lot: 101), the back-calculated concentration of the QC-LLOQ sample was 129.5 % and therefore did not fulfill the acceptance criteria of ± 20 % of the nominal concentration. In the blank sample of this matrix lot, no signal at the retention time of 2439-exNA was detected. The selectivity test of this matrix lot was repeated in the two following validation batches: AB02 and AB03 (according to KU-RGX-SOP-005). In the

repetitions, the back-calculated concentration was 98.0 % in AB02 and 117.5 % in AB03. In the blank samples no signal at the retention time of 2439-exNA was detected. Therefore, the matrix lot 101 fulfilled the acceptance criteria regarding the back-calculated concentration and the analyte response in the blank samples of the matrix lot.

- For all other brain matrix lots (Lot: 102, 103, 151, 152,153), no signal at the retention time of 2439-exNA was detected in the selectivity blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same matrix lot.
- The back-calculated concentrations of the QC-LLOQ samples ranged from 94.0 % – 104.0 % of the nominal concentration.

Data are listed in Table 11.

Results for Kidney Tissue:

- No signal at the retention time of 2439-exNA was detected in the selectivity blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same matrix lot.
- The back-calculated concentrations of the QC-LLOQ samples ranged from 89.0 % – 104.5 % of the nominal concentration.

Data are listed in Table 12.

Results for Liver Tissue:

- In 1 of 6 selectivity blank samples (Lot: 153), a signal at the retention time of 2439-exNA was detected. The signal was higher than one fifth of the 2439-exNA response (peak area) in the corresponding QC-LLOQ sample of the same matrix lot. The selectivity test of these matrix lot was repeated in the two following validation batches AB02 and AB03 (according to KU-RGX-SOP-005). In both repetitions, no signal at the retention time of 2439-exNA was detected in the selectivity blank samples and hence, the 2439-exNA response (peak area) in the LLOQ calibration standards was at least 5 times higher than the response (peak area) in the blank sample of the matrix lot.
- In all other selectivity blank samples (Lot: 101, 102, 103, 151, 152), no signal at the retention time of 2439-exNA was detected in the selectivity blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same matrix lot.
- The back-calculated concentrations of the QC-LLOQ samples ranged from 87.0 % – 106.0 % of the nominal concentration.

Data are listed in Table 13.

Selectivity of the method was confirmed for dog tissue.

9.4 Matrix Effect

The matrix effect was evaluated by analyzing triplicates of QC-Low and QC-High samples, each prepared using 6 different blank dog tissue lots.

Acceptance Criteria:

For each evaluated blank dog tissue lot, the mean back-calculated concentration of the QC-Low and QC-High samples (each triplicates) should be within ± 15 % of the nominal concentrations and the CV should not exceed 15 %.

Results for Brain Tissue:

- For 6 matrix lots (101, 102, 103, 151, 152, 153) the mean back-calculated concentrations of the QC samples ranged from 88.2 % – 97.8 % (QC-Low) and 95.9 % – 102.3 % (QC-High) of the nominal concentrations, and CV values ranged from 0.8 % – 3.4 % (QC-Low) and 0.0 % – 0.7 % (QC-High).

Data are listed in Table 14.

Results for Kidney Tissue:

- For 6 matrix lots (101, 102, 103, 151, 152, 153), the mean back-calculated concentrations of the QC samples ranged from 89.0 % – 96.2 % (QC-Low) and 95.7 % – 100.7 % (QC-High) of the nominal concentrations, and CV values ranged from 1.4 % – 5.0 % (QC-Low) and 0.1 % – 1.1 % (QC-High).

Data are listed in Table 15.

Results for Liver Tissue:

- For 6 matrix lots (101, 102, 103, 151, 152, 153), the mean back-calculated concentrations of the QC samples ranged from 91.6 % – 96.4 % (QC-Low) and 96.2 % – 101.7 % (QC-High) of the nominal concentrations, and CV values ranged from 0.5 % – 5.0 % (QC-Low) and 0.1 % – 0.9 % (QC-High).

Data are listed in Table 16.

Based on these results, there is no matrix effect for the assay regarding the analysis of 2439-exNA in dog tissue.

9.5 Carry-Over

Blank samples were injected after the ULOQ calibration standards to assess carry-over.

Acceptance Criteria:

The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the analyte response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Results:

- No signal for 2439-exNA was detected in the carry-over blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Data are listed in Table 17.

No carry-over was observed for 2439-exNA in dog tissue. Therefore, no special precautions after the analysis of samples with a high concentration of 2439-exNA have to be implemented.

9.6 Quality Control (QC) Samples

Analysis of assay accuracy and precision, and stability was conducted by analysis of QC samples of defined concentrations.

QC samples were prepared by spiking known amounts of 2439-exNA into dog brain, liver and kidney lysates (and homogenates where necessary).

QC samples were prepared at 4 concentration levels which cover the calibration curve range:

- at the LLOQ (QC-LLOQ): 20.0 ng/g
- at 3 times above the LLOQ (QC-Low): 60.0 ng/g
- at 30 – 50 % of the calibration curve range (QC-Mid): 4,500.0 ng/g
- at least at 75 % of the upper calibration curve range (QC-High): 9,000.0 ng/g

For evaluation of dilution integrity one additional QC concentration level was prepared:

- at 500 times above the ULOQ (QC-500xULOQ): 5,000,000.0 ng/g

9.7 Accuracy

The accuracy of the analytical method describes the closeness of the determined concentration obtained by the method to the nominal concentration of 2439-exNA (expressed in percentage).

The analysis of assay accuracy was conducted using the QC samples at the 4 concentration levels that were measured as 6 individually prepared replicates per analytical run.

The analysis of assay accuracy was divided into within-run and between-run accuracy. The within-run accuracy was repeated 3 times, prepared by 2 technicians on different days, and analyzed on 2 independent HPLC systems. Thus, 3 independent data sets for the between-run accuracy were generated.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate for each data set.

9.7.1 Within-Run Accuracy

Acceptance Criteria:

The mean back-calculated concentration of the QC samples at each of the 4 concentration levels and in each of the 3 data sets should be within ± 15 % of the nominal concentrations (± 20 % for the QC-LLOQ).

Results for Brain Tissue:

- The mean back-calculated concentrations of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 102.3 % – 106.9 % (QC-LLOQ), 89.4 % – 96.5 % (QC-Low), 96.6 % – 101.5 % (QC-Mid) and 96.1 % – 102.0 % (QC-High) of the nominal concentrations.

Data are listed in Table 18.

Results for Kidney Tissue:

- The mean back-calculated concentrations of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 93.6 % – 104.2 % (QC-LLOQ), 90.5 % – 94.9 % (QC-Low), 96.9 % – 100.5 % (QC-Mid) and 95.7 % – 100.5 % (QC-High) of the nominal concentrations.

Data are listed in Table 19.

Results for Liver Tissue:

- The mean back-calculated concentrations of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 92.8 % – 107.0 % (QC-LLOQ), 93.5 % – 95.6 % (QC-Low), 97.2 % – 101.0 % (QC-Mid) and 96.4 % – 101.5 % (QC-High) of the nominal concentrations.

Data are listed in Table 20.

9.7.2 Between-Run Accuracy

Acceptance Criteria:

The mean back-calculated concentrations of all QC samples at each of the 4 concentration levels of all 3 data sets should be within ± 15 % of the nominal concentrations (± 20 % for the QC-LLOQ).

Results for Brain Tissue:

- The mean back-calculated concentrations of all QC samples at each of 4 concentration levels of all 3 data sets were 104.3 % (QC-LLOQ), 93.7 % (QC-Low), 99.4 % (QC-Mid), and 99.6 % (QC-High) of the nominal concentrations.

Data are listed in Table 18.

Results for Kidney Tissue:

- The mean back-calculated concentrations of all QC samples at each of 4 concentration levels of all 3 data sets were 99.7 % (QC-LLOQ), 93.1 % (QC-Low), 99.2 % (QC-Mid), and 98.9 % (QC-High) of the nominal concentrations.

Data are listed in Table 19.

Results for Liver Tissue:

- The mean back-calculated concentrations of all QC samples at each of 4 concentration levels of all 3 data sets were 101.2 % (QC-LLOQ), 94.7 % (QC-Low), 99.3 % (QC-Mid), and 99.2 % (QC-High) of the nominal concentrations.

Data are listed in Table 20.

Based on these results, the assay for 2439-exNA in dog tissue is considered accurate.

9.8 Precision

The precision of the analytical method describes the closeness of repeated individual measures of the 2439-exNA concentration, expressed as coefficient of variation (CV).

The analysis of assay precision was conducted using the QC samples at the 4 concentration levels. All concentration levels were measured as 6 individually prepared replicates per analytical run.

As for accuracy, precision was demonstrated within a single analytical run and between different analytical runs. The same analytical runs and data were used for the demonstration of accuracy.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate for each data set.

9.8.1 Within-Run Precision

Acceptance Criteria:

The within-run CV of the QC samples at each of the 4 concentrations levels and in each of the 3 data sets should not exceed 15 % (20 % for the QC-LLOQ).

Results for Brain Tissue:

- The within-run CV of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 5.0 % – 13.5 % (QC-LLOQ), 1.5 % – 3.3 % (QC-Low), 0.5 % – 0.9 % (QC-Mid) and 0.3 % – 0.6 % (QC-High).

Data are listed in Table 18.

Results for Kidney Tissue:

- The within-run CV of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 6.4 % – 8.9 % (QC-LLOQ), 2.9 % – 3.9 % (QC-Low), 0.4 % – 0.8 % (QC-Mid) and 0.1 % – 1.0 % (QC-High).

Data are listed in Table 19.

Results for Liver Tissue:

- The within-run CV of the QC samples at each of the 4 concentration levels in the 3 data sets ranged from 4.7 % – 6.5 % (QC-LLOQ), 1.8 % – 4.1 % (QC-Low), 0.4 % – 0.9 % (QC-Mid) and 0.4 % – 0.6 % (QC-High).

Data are listed in Table 20.

9.8.2 Between-Run Precision

Between-run precision was determined with an identical setup as for within-run precision but included the assessment of QC samples from 3 independent data sets prepared by 2 technicians on 3 different days on 2 independent HPLC systems.

Acceptance Criteria:

The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets should not exceed 15 % (20 % for the QC-LLOQ).

Results for Brain Tissue:

- The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets were 8.6 % (QC-LLOQ), 4.1 % (QC-Low), 2.2 % (QC-Mid), and 2.6 % (QC-High).

Data are listed in Table 18.

Results for Kidney Tissue:

- The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets were 8.8 % (QC-LLOQ), 3.7 % (QC-Low), 1.8 % (QC-Mid), and 2.4 % (QC-High).

Data are listed in Table 19.

Results for Liver Tissue:

- The between-run CV of all QC samples at each of the 4 concentration levels of all 3 data sets were 8.1 % (QC-LLOQ), 2.9 % (QC-Low), 1.8 % (QC-Mid), and 2.2 % (QC-High).

Data are listed in Table 20.

Based on these results, the assay for 2439-exNA in dog tissue is considered precise.

9.9 Dilution Integrity

Dilution integrity was evaluated using QC samples prepared at the concentration level 500-fold (5,000,000.0 ng/g; QC-500xULOQ) above the ULOQ (6 replicates for each dilution factor).

The 5,000,000.0 ng/g QC samples were analyzed and quantified after a 1:1,000 dilution.

The QC samples were diluted with buffer solution (10 % ACN / 1x PBS / 1 mM TCEP).

Acceptance Criteria

For each evaluated dilution factor and concentration, the mean back-calculated concentration of the QC samples should be within ± 15 % of the nominal concentration and the CV should not exceed 15 %.

Results:

- The mean back-calculated concentration was 86.2 % for the 5,000,000.0 ng/g QC samples (1:1,000 dilution).
- The CV was 12.1 % for the 5,000,000.0 ng/g QC samples (1:1,000 dilution).

Data are listed in Table 21.

Dilution of samples up to 1:1,000 up to a concentration of 5,000,000.0 ng/g met the acceptance criteria for precision and accuracy.

9.10 Stability

The stability of 2439-exNA was analyzed under several conditions relevant to the assay setup.

2439-exNA was spiked at 2 concentration levels into dog brain homogenate and / or lysate samples (depending on the test) and analyzed in triplicate. The concentration levels were identical to the QC-Low and QC-High samples, with concentrations of 60.0 ng/g and 9,000.0 ng/g. Calibration curves were freshly prepared for each analytical run.

Long-term stability, freeze-thaw stability, and short-term stability were evaluated in polypropylene tubes (Eppendorf Protein LoBind tubes).

Long-term stability of 2439-exNA in dog tissue will be assessed in brain homogenates (preparation excludes Proteinase K), which cover study sample stability from tissue preparation until tissue lysis and in brain lysates (preparation includes Proteinase K), which cover study sample stability after tissue lysis.

Freeze-thaw stability and bench-top stability/robustness of 2439-exNA in dog brain were assessed in tissue lysates only.

9.10.1 Long-Term Stability

Aliquots of 2439-exNA samples in dog brain homogenates at the 2 QC concentration levels were stored at -80 °C ± 20 °C, and were and will be analyzed at up to 2 time points (T= 0 and 4.5 months) in triplicate.

Aliquots of 2439-exNA samples in dog brain lysates at the 2 QC concentration levels were stored at -80 °C ± 20 °C, and were and will be analyzed at up to 2 time points (T= 0 and 3 months) in triplicate.

Acceptance Criteria:

For each evaluated condition, the mean back-calculated concentration of the QC-Low and QC-High samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results of Homogenates:

At the time the Method Validation Report was prepared, analysis of the first time point (T= 0) was available.

- At T= 0, the mean back-calculated concentrations were 93.1 % (QC-Low) and 100.2 % (QC-High) of the nominal concentrations.

Data are listed in Table 22.

Results of Lysates:

At the time the Method Validation Report was prepared, analysis of the first time point (T= 0) was available.

- At T= 0, the mean back-calculated concentrations were 97.3 % (QC-Low) and 100.1 % (QC-High) of the nominal concentrations.

Data are listed in Table 23.

Long-term stability of 2439-exNA in dog brain homogenates will be further monitored for up to 4.5 months and in dog brain lysates for up to 3 months. The final long-term stability data set will be added as an addendum to the Method Validation Report at the end of the study.

9.10.2 Freeze-Thaw and Short-Term Stability

Aliquots of 2439-exNA samples in dog brain lysates at the 2 QC concentration levels were stored at -80 °C ± 20 °C for at least 12 hours.

Thawing of the QC samples from -80 °C ± 20 °C was done on the bench at room temperature. After thawing, the QC samples were stored on the bench at room temperature for 4 hours before further processing.

Subsequently, the QC samples were refrozen for at least 12 hours at -80 °C ± 20 °C to start the next freeze-thaw cycle and to repeat the process described above.

After 5 freeze-thaw cycles, the QC samples were processed and analyzed in triplicate according to the assay conditions.

Acceptance Criteria:

The mean back-calculated concentration of the QC-Low and QC-High samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results:

- After 5 freeze/thaw cycles with 4 hours at room temperature, the mean back-calculated concentrations were 102.2 % (QC-Low) and 101.2 % (QC-High) of the nominal concentrations.

Data are listed in Table 24.

At the evaluated concentrations, 2439-exNA is stable in dog brain lysates when thawed up to 5 times at room temperature during sample analysis and for up to 4 hours during sample storage at room temperature.

9.10.3 Short-Term Stability

Short-term stability of 2439-exNA in dog brain could not be directly tested as the analyte will be digested by nucleases present in the biological matrix upon thawing. Therefore, short-term stability was assessed after the Proteinase K treatment during bench-top stability/robustness testing (Section 9.10.4).

9.10.4 Bench-Top Stability/Robustness

Bench-top stability/robustness of 2439-exNA was assessed under prolonged treatment during one critical step of the assay.

Aliquots of 2439-exNA samples in dog brain at the 2 QC concentration levels were stored for 2 hours on the bench at room temperature after the Proteinase K treatment and before the sodium dodecyl sulfate (SDS) precipitation with potassium chloride (KCl). After the SDS precipitation and before the hybridization procedure, the QC samples were stored for another 2 hours on the bench at room temperature.

Acceptance Criteria:

The mean back-calculated concentration of the QC-Low and QC-High samples (each triplicates) should be within ± 15 % of the nominal concentrations.

Results:

- After storage for 2 hours before and 2 hours after the SDS precipitation step, the mean back-calculated concentrations were 92.6 % (QC-Low) and 97.5 % (QC-High) of the nominal concentrations.

Data are listed in Table 25.

At the evaluated concentrations, 2439-exNA in dog brain lysates is stable during prolonged storage on the bench at room temperature for up to 2 hours before and for up to 2 hours after the SDS precipitation step.

9.10.5 Stability of the Processed Sample/On-Instrument Stability (Autosampler)

Stability of the processed sample of 2439-exNA in dog brain lysates was assessed with 4 QC sets of QC-Low and QC-High triplicates that were stored on the HPLC autosampler at 20 °C ± 3 °C and were analyzed at different time points, covering 119 hours between the analysis of the first and the last QC set.

Acceptance Criteria:

For each evaluated time point, the mean back-calculated concentration of the QC-Low and QC-High samples should be within ± 15 % of the nominal concentrations.

The QC set of last time point, which fulfills the acceptance criteria, defines processed sample stability.

Results:

- After time point 1, the mean back-calculated concentrations were 93.3 % (QC-Low) and 103.2 % (QC-High) of the nominal concentrations.
- After time point 2 (covering 37 hours), the mean back-calculated concentrations were 91.2 % (QC-Low) and 102.6 % (QC-High) of the nominal concentrations.
- After time point 3 (covering 46 hours), the mean back-calculated concentrations were 96.8 % (QC-Low) and 104.5 % (QC-High) of the nominal concentrations.
- After time point 4 (covering 119 hours), the mean back-calculated concentrations were 93.8 % (QC-Low) and 103.2 % (QC-High) of the nominal concentrations.

Data are listed in Table 26.

At the evaluated concentrations, 2439-exNA in processed dog brain lysate samples is stable during storage on the HPLC autosampler at 20 °C ± 3 °C for up to 119 hours.

If a longer period of sample storage on the autosampler should be necessary, the last QC sample set and calibration curve will verify the validity of the analysis.

9.10.6 Stability of Stock Solution and Working Solutions

9.10.6.1 Short-Term Stability of Stock Solution and Working Solutions

Aliquots of the lowest concentration of the working solution (working solution LLOQ) of 2439-exNA were stored at room temperature for 4 hours and analyzed in triplicate.

The samples were analyzed by comparing the instrument's response with a freshly prepared working solution of the same concentration.

Acceptance Criteria:

The mean back-calculated concentration of the working solution LLOQ (each triplicates) should be within ± 20 % of the nominal concentration.

Results:

- After 4 hours at room temperature, the mean back-calculated concentration was 94.2 % (working solution LLOQ) of the nominal concentration.

Data are listed in Table 27.

At the evaluated concentration, the lowest concentration of the working solution of 2439-exNA is stable during storage for up to 4 hours at room temperature.

Short-term stability of stock solution and highest concentration of the working solution was not performed, as the stability test of the stock solution and the highest concentration of the working solution bracketing the concentration used herein, was already conducted in the method validation for the analysis of 2439-exNA in dog plasma (Axolabs Project: AN402.0-24).

9.10.6.2 Long-Term Stability of Stock Solution

Long-term stability of the stock solution was not performed, as stability data are provided by the Sponsor.

10 Deviations

The bioanalytical method validation for the analysis of 2439-exNA in dog tissue was conducted according to the Method Validation Plan and to applicable SOPs without any deviation.

11 Conclusions

The bioanalytical method for the analysis of 2439-exNA in dog tissue was successfully validated and is described in the STM *STM_Broad_2439-exNA_DogTissue* (edition 02, effective date: 03-Dec-2024). All validation tests fulfilled the acceptance criteria described in the Method Validation Plan.

The method can be used within a linear range from 20.0 ng/g to 10,000.0 ng/g of 2439-exNA in dog tissue, applying a 1/X weighted linear regression model.

2439-exNA is stable in dog brain lysates at -80 °C ± 20 °C during up to 5 freeze/thaw cycles.

The evaluation of long-term stability of 2439-exNA in dog brain homogenates and lysates is still ongoing, and the results will be added as an addendum to this Method Validation Report after the finalization of the long-term stability testing.

12 List of Records to be Retained

All original data and documents generated during this method validation will be archived at Axolabs GmbH under appropriate conditions for 4 Years.

The following records will be retained:

- Final signed Method Validation Plan
- Critical communication, as appropriate
- Raw data:
    - HPLC file(s) with HPLC chromatograms and HPLC program
    - Assay protocols
    - System Suitability Test reports
    - CoA (or equivalent) of PNA probe and reference standard
    - Other raw data (if applicable)
- QC records of document and data reviews/audits
- Final signed Method Validation Report and all addenda

A backup file of all analytical HPLC data, together with an electronic PDF report of all analytical runs, and all electronic files generated during the method validation (reports, Excel-Files) are stored in electronic form.

13 Data Tables

Table 7: Analytical Batches for the Analysis of 2439-exNA in Dog Tissue

Run ID	Date of Analysis	Description	Status	Comments
AN406.0-24_01_AB01	14-Nov-2024	1st main validation batch	Passed	n.a.
AN406.0-24_01_AB02	20-Nov-2024	2nd main validation batch	Passed	n.a.
AN406.0-24_01_AB03	22-Nov-2024	3rd main validation batch	Passed	n.a.

Table 8: Calibration Curve Parameters for the Analysis of 2439-exNA in Dog Brain

Equation for the calibration curve using a linear regression model weighted 1/X: y = m*x + b					
Run ID	**Date of Analysis**	**Sample No.**	**Slope [m]**	**Y-Offset [b]**	**R-Squared**
AN406.0-24_01_AB01	14-Nov-2024	6 - 11 and 145 - 150	0.00206	-0.00089	0.99997
AN406.0-24_01_AB02	20-Nov-2024	6 - 11 and 126 - 131	0.00149	-0.00261	0.99998
AN406.0-24_01_AB03	22-Nov-2024	4 - 9 and 126 - 131	0.00146	-0.00372	0.99999

Calibration standards that did not fulfill the acceptance criteria and blank samples were excluded from the calculation of the calibration curves.

Table 9: Results of the Calibration Standards for the Analysis of 2439-exNA in Dog Brain

		Cal-Level-2 (LLOQ)		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7 (ULOQ)	
		20.0 ng/g		100.0 ng/g		500.0 ng/g		1,000.0 ng/g		5,000.0 ng/g		10,000.0 ng/g	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB01	6 - 11	20.4	102.0 %	97.5	97.5 %	506.5	101.3 %	1,024.4	102.4 %	5,010.8	100.2 %	9,997.1	100.0 %
AB01	145 - 150	19.3	96.5 %	98.1	98.1 %	501.4	100.3 %	1,030.2	103.0 %	4,956.4	99.1 %	9,978.0	99.8 %
AB02	6 - 11	21.1	105.5 %	95.3	95.3 %	496.8	99.4 %	1,015.7	101.6 %	5,016.1	100.3 %	9,974.1	99.7 %
AB02	126 - 131	20.2	101.0 %	96.2	96.2 %	499.7	99.9 %	1,003.6	100.4 %	5,036.2	100.7 %	9,964.8	99.6 %
AB03	4 - 9	21.4	107.0 %	98.9	98.9 %	493.5	98.7 %	1,000.9	100.1 %	4,975.4	99.5 %	10,050.3	100.5 %
AB03	126 - 131	20.2	101.0 %	97.0	97.0 %	488.8	97.8 %	997.2	99.7 %	4,988.9	99.8 %	10,007.6	100.1 %
	Mean	20.43	102.2 %	97.17	97.2 %	497.78	99.6 %	1,012.00	101.2 %	4,997.30	99.9 %	9,995.32	100.0 %
	SD (±)	0.74		1.30		6.21		13.50		29.23		31.17	
	CV (%)	3.6 %		1.3 %		1.2 %		1.3 %		0.6 %		0.3 %	
	N	6		6		6		6		6		6	

Table 10: Sensitivity/Lower Limit of Quantification (LLOQ) for the Analysis of 2439-exNA in Dog Brain

		Cal-Level-1 (Blank Sample)		Cal-Level-2 (LLOQ)	
		0.0 ng/g		20.0 ng/g	
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/g**
AB01	5, 6	0.0000	0.0 %	0.0410	20.4
AB01	144, 145	0.0000	0.0 %	0.0388	19.3
AB02	5, 6	0.0000	0.0 %	0.0290	21.1
AB02	125, 126	0.0000	0.0 %	0.0276	20.2
AB03	3, 4	0.0000	0.0 %	0.0277	21.4
AB03	125, 126	0.0000	0.0 %	0.0258	20.2

Table 11: Selectivity for the Analysis of 2439-exNA in Dog Brain

					Blank Sample for Selectivity		QC-LLOQ		
					0.0 ng/g		20.0 ng/g		
Run ID	**Date of Analysis**	**Sex**	**Blank Matrix Lot No.**	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**
AB01	14-Nov-2024	male	101	28, 35[2)]	0.0000	0.0 %	0.0524[2)]	25.9[2)]	129.5 %[2)]
		male	102	29, 36	0.0000	0.0 %	0.0418	20.8	104.0 %
		male	103	30, 37	0.0000	0.0 %	0.0382	19.0	95.0 %
		female	151	31, 38	0.0000	0.0 %	0.0388	19.3	96.5 %
		female	152	32, 39	0.0000	0.0 %	0.0380	18.9	94.5 %
		female	153	33, 40	0.0000	0.0 %	0.0378	18.8	94.0 %
AB02	20-Nov-2024	male	101	105, 106	0.0000	0.0 %	0.0267	19.6	98.0 %
AB03	22-Nov-2024	male	101	110, 111	0.0000	0.0 %	0.0307	23.5	117.5 %

[2)] The marked and underlined sample did not fulfill the acceptance criterion. Therefore, the selectivity test was repeated for the corresponding lot in subsequent batches.

Table 12: Selectivity for the Analysis of 2439-exNA in Dog Kidney

					Blank Sample for Selectivity		QC-LLOQ		
					0.0 ng/g		20.0 ng/g		
Run ID	**Date of Analysis**	**Sex**	**Blank Matrix Lot No.**	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**
AB01	14-Nov-2024	male	101	98, 105	0.0000	0.0 %	0.0360	17.9	89.5 %
		male	102	99, 106	0.0000	0.0 %	0.0360	17.9	89.5 %
		male	103	100, 107	0.0000	0.0 %	0.0358	17.8	89.0 %
		male	104	101, 108	0.0000	0.0 %	0.0421	20.9	104.5 %
		male	105	102, 109	0.0000	0.0 %	0.0387	19.2	96.0 %
		male	106	103, 110	0.0000	0.0 %	0.0374	18.6	93.0 %

Table 13: Selectivity for the Analysis of 2439-exNA in Dog Liver

					Blank Sample for Selectivity		QC-LLOQ		
					0.0 ng/g		20.0 ng/g		
Run ID	**Date of Analysis**	**Sex**	**Blank Matrix Lot No.**	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**
AB01	14-Nov-2024	male	101	63, 70	0.0000	0.0 %	0.0373	18.6	93.0 %
		male	102	64, 71	0.0000	0.0 %	0.0357	17.8	89.0 %
		male	103	65, 72	0.0000	0.0 %	0.0369	18.4	92.0 %
		female	151	66, 73	0.0000	0.0 %	0.0350	17.4	87.0 %
		female	152	67, 74	0.0000	0.0 %	0.0411	20.4	102.0 %
		female	153	68[2)], 75	0.0624[2)]	165.1 %[2)]	0.0378	18.8	94.0 %
AB02	20-Nov-2024	female	153	108, 109	0.0000	0.0 %	0.0290	21.2	106.0 %
AB03	22-Nov-2024	female	153	113, 114	0.0000	0.0 %	0.0280	21.7	108.5 %

[2)] The marked and underlined sample did not fulfill the acceptance criterion. Therefore, the selectivity test was repeated for the corresponding lot in subsequent batches.

Table 14: Matrix Effect for the Analysis of 2439-exNA in Dog Brain

			QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB01	101	42, 56	51.8	86.3 %	8,695.1	96.6 %
		43, 57	53.7	89.5 %	8,667.7	96.3 %
		44, 58	53.2	88.7 %	8,666.6	96.3 %
		Mean	52.90	88.2 %	8,676.47	96.4 %
		SD (±)	0.98		16.15	
		CV (%)	1.9 %		0.2 %	
		N	3		3	
AB01	102	45, 59	53.5	89.2 %	8,614.5	95.7 %
		46, 60	56.1	93.5 %	8,642.4	96.0 %
		47, 61	53.6	89.3 %	8,624.9	95.8 %
		Mean	54.40	90.7 %	8,627.27	95.9 %
		SD (±)	1.47		14.10	
		CV (%)	2.7 %		0.2 %	
		N	3		3	

Table 14: Matrix Effect for the Analysis of 2439-exNA in Dog Brain (Continued)

			QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB02	103	28, 42	57.3	95.5 %	9,177.3	102.0 %
		29, 43	58.2	97.0 %	9,279.7	103.1 %
		30, 44	57.5	95.8 %	9,166.6	101.9 %
		Mean	57.67	96.1 %	9,207.87	102.3 %
		SD (±)	0.47		62.44	
		CV (%)	0.8 %		0.7 %	
		N	3		3	
AB02	151	31, 45	56.8	94.7 %	9,134.9	101.5 %
		32, 46	57.0	95.0 %	9,209.4	102.3 %
		33, 47	55.6	92.7 %	9,127.2	101.4 %
		Mean	56.47	94.1 %	9,157.17	101.7 %
		SD (±)	0.76		45.40	
		CV (%)	1.3 %		0.5 %	
		N	3		3	

Table 14: Matrix Effect for the Analysis of 2439-exNA in Dog Brain (Continued)

			QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB03	152	26, 40	55.0	91.7 %	9,069.6	100.8 %
		27, 41	58.1	96.8 %	9,126.2	101.4 %
		28, 42	58.1	96.8 %	9,062.4	100.7 %
		Mean	57.07	95.1 %	9,086.07	101.0 %
		SD (±)	1.79		34.94	
		CV (%)	3.1 %		0.4 %	
		N	3		3	
AB03	153	29, 43	57.3	95.5 %	9,050.1	100.6 %
		30, 44	61.0	101.7 %	9,047.6	100.5 %
		31, 45	57.8	96.3 %	9,045.3	100.5 %
		Mean	58.70	97.8 %	9,047.67	100.5 %
		SD (±)	2.01		2.40	
		CV (%)	3.4 %		0.0 %	
		N	3		3	

Table 15: Matrix Effect for the Analysis of 2439-exNA in Dog Kidney

			QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB01	101	112, 126	56.6	94.3 %	8,616.5	95.7 %
		113, 127	55.8	93.0 %	8,614.6	95.7 %
		114, 128	53.3	88.8 %	8,626.5	95.9 %
		Mean	55.23	92.1 %	8,619.20	95.8 %
		SD (±)	1.72		6.39	
		CV (%)	3.1 %		0.1 %	
		N	3		3	
AB01	102	115, 129	53.5	89.2 %	8,618.3	95.8 %
		116, 130	54.1	90.2 %	8,620.1	95.8 %
		117, 131	52.6	87.7 %	8,597.8	95.5 %
		Mean	53.40	89.0 %	8,612.07	95.7 %
		SD (±)	0.75		12.39	
		CV (%)	1.4 %		0.1 %	
		N	3		3	

Table 15: Matrix Effect for the Analysis of 2439-exNA in Dog Kidney (Continued)

			QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB02	103	84, 98	55.1	91.8 %	9,102.0	101.1 %
		85, 99	59.2	98.7 %	8,933.6	99.3 %
		86, 100	54.7	91.2 %	9,099.9	101.1 %
		Mean	56.33	93.9 %	9,045.17	100.5 %
		SD (±)	2.49		96.63	
		CV (%)	4.4 %		1.1 %	
		N	3		3	
AB02	151	87, 101	56.3	93.8 %	8,949.6	99.4 %
		88, 102	55.6	92.7 %	9,058.2	100.6 %
		89, 103	57.2	95.3 %	9,145.4	101.6 %
		Mean	56.37	93.9 %	9,051.07	100.6 %
		SD (±)	0.80		98.09	
		CV (%)	1.4 %		1.1 %	
		N	3		3	

Table 15: Matrix Effect for the Analysis of 2439-exNA in Dog Kidney (Continued)

			QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB03	152	82, 96	54.0	90.0 %	9,086.6	101.0 %
		83, 97	59.3	98.8 %	9,103.8	101.2 %
		84, 98	55.0	91.7 %	8,990.8	99.9 %
		Mean	56.10	93.5 %	9,060.40	100.7 %
		SD (±)	2.82		60.89	
		CV (%)	5.0 %		0.7 %	
		N	3		3	
AB03	153	85, 99	56.8	94.7 %	8,979.9	99.8 %
		86, 100	57.0	95.0 %	9,013.4	100.1 %
		87, 101	59.4	99.0 %	9,000.0	100.0 %
		Mean	57.73	96.2 %	8,997.77	100.0 %
		SD (±)	1.45		16.86	
		CV (%)	2.5 %		0.2 %	
		N	3		3	

Table 16: Matrix Effect for the Analysis of 2439-exNA in Dog Liver

			QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB01	101	77, 91	54.7	91.2 %	8,626.6	95.9 %
		78, 92	56.7	94.5 %	8,679.9	96.4 %
		79, 93	60.4	100.7 %	8,668.8	96.3 %
		Mean	57.27	95.4 %	8,658.43	96.2 %
		SD (±)	2.89		28.12	
		CV (%)	5.0 %		0.3 %	
		N	3		3	
AB01	102	80, 94	54.6	91.0 %	8,728.3	97.0 %
		81, 95	56.2	93.7 %	8,683.5	96.5 %
		82, 96	54.1	90.2 %	8,692.7	96.6 %
		Mean	54.97	91.6 %	8,701.50	96.7 %
		SD (±)	1.10		23.66	
		CV (%)	2.0 %		0.3 %	
		N	3		3	

Table 16: Matrix Effect for the Analysis of 2439-exNA in Dog Liver (Continued)

			QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB02	103	56, 70	56.2	93.7 %	9,174.8	101.9 %
		57, 71	58.2	97.0 %	9,187.0	102.1 %
		58, 72	59.1	98.5 %	9,093.0	101.0 %
		Mean	57.83	96.4 %	9,151.60	101.7 %
		SD (±)	1.48		51.11	
		CV (%)	2.6 %		0.6 %	
		N	3		3	
AB02	151	59, 73	56.0	93.3 %	9,125.9	101.4 %
		60, 74	56.4	94.0 %	9,100.3	101.1 %
		61, 75	55.8	93.0 %	9,138.0	101.5 %
		Mean	56.07	93.4 %	9,121.40	101.3 %
		SD (±)	0.31		19.25	
		CV (%)	0.5 %		0.2 %	
		N	3		3	

Table 16: Matrix Effect for the Analysis of 2439-exNA in Dog Liver (Continued)

			QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB03	152	54, 68	57.5	95.8 %	8,976.2	99.7 %
		55, 69	58.6	97.7 %	8,994.3	99.9 %
		56, 70	56.2	93.7 %	8,972.9	99.7 %
		Mean	57.43	95.7 %	8,981.13	99.8 %
		SD (±)	1.20		11.52	
		CV (%)	2.1 %		0.1 %	
		N	3		3	
AB03	153	57, 71	58.5	97.5 %	8,863.9	98.5 %
		58, 72	56.8	94.7 %	8,973.0	99.7 %
		59, 73	56.5	94.2 %	9,026.1	100.3 %
		Mean	57.27	95.4 %	8,954.33	99.5 %
		SD (±)	1.08		82.70	
		CV (%)	1.9 %		0.9 %	
		N	3		3	

Table 17: Carry-Over for the Analysis of 2439-exNA in Dog Brain

		Cal-Level-2 (LLOQ)			Cal-Level-7 (ULOQ)			Blank Sample for Carry-Over		
		20.0 ng/g			10,000.0 ng/g			0.0 ng/g		
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**	**Peak Area [mv*min]**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ≤ 20 % of LLOQ**
AB01	6, 11, 12	0.0410	20.4	102.0 %	20.5609	9,997.1	100.0 %	0.0000	BQL	0.0 %
AB01	145, 150, 151	0.0388	19.3	96.5 %	20.5215	9,978.0	99.8 %	0.0000	BQL	0.0 %
AB02	6, 11, 12	0.0290	21.1	105.5 %	14.9052	9,974.1	99.7 %	0.0000	BQL	0.0 %
AB02	126, 131, 132	0.0276	20.2	101.0 %	14.8912	9,964.8	99.6 %	0.0000	BQL	0.0 %
AB03	4, 9, 10	0.0277	21.4	107.0 %	14.7110	10,050.3	100.5 %	0.0000	BQL	0.0 %
AB03	126, 131, 132	0.0258	20.2	101.0 %	14.6485	10,007.6	100.1 %	0.0000	BQL	0.0 %

Table 18: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Dog Brain

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					20.0 ng/g		60.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ±15 % of Nominal**
AB01	14-Nov-2024	LJ	HPLC-GLP-11	35, 42, 49, 56	25.9	129.5 %	51.8	86.3 %	4,344.7	96.5 %	8,695.1	96.6 %
				36, 43, 50, 57	20.8	104.0 %	53.7	89.5 %	4,347.5	96.6 %	8,667.7	96.3 %
				37, 44, 51, 58	19.0	95.0 %	53.2	88.7 %	4,423.4	98.3 %	8,666.6	96.3 %
				38, 45, 52, 59	19.3	96.5 %	53.5	89.2 %	4,319.4	96.0 %	8,614.5	95.7 %
				39, 46, 53, 60	18.9	94.5 %	56.1	93.5 %	4,319.5	96.0 %	8,642.4	96.0 %
				40, 47, 54, 61	18.8	94.0 %	53.6	89.3 %	4,317.8	96.0 %	8,624.9	95.8 %
Within-Run Accuracy/Precision I				**Mean**	20.45	102.3 %	53.65	89.4 %	4,345.38	96.6 %	8,651.87	96.1 %
				SD (±)	2.77		1.39		40.49		30.17	
				CV (%)	13.5 %		2.6 %		0.9 %		0.3 %	
				N	6		6		6		6	
AB02	20-Nov-2024	TKu	HPLC-GLP-9	21, 28, 35, 42	21.6	108.0 %	57.3	95.5 %	4,600.8	102.2 %	9,177.3	102.0 %
				22, 29, 36, 43	21.4	107.0 %	58.2	97.0 %	4,585.4	101.9 %	9,279.7	103.1 %
				23, 30, 37, 44	22.6	113.0 %	57.5	95.8 %	4,549.4	101.1 %	9,166.6	101.9 %
				24, 31, 38, 45	20.9	104.5 %	56.8	94.7 %	4,551.9	101.2 %	9,134.9	101.5 %
				25, 32, 39, 46	19.1	95.5 %	57.0	95.0 %	4,567.4	101.5 %	9,209.4	102.3 %
				26, 33, 40, 47	22.7	113.5 %	55.6	92.7 %	4,537.1	100.8 %	9,127.2	101.4 %
Within-Run Accuracy/Precision II				**Mean**	21.38	106.9 %	57.07	95.1 %	4,565.33	101.5 %	9,182.52	102.0 %
				SD (±)	1.32		0.87		24.07		56.17	
				CV (%)	6.2 %		1.5 %		0.5 %		0.6 %	
				N	6		6		6		6	

Underlined QC sample did not fulfill the acceptance criterion but was included in the statistical calculations (Mean, SD and CV).

Table 16: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Dog Brain (Continued)

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					20.0 ng/g		60.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ±15 % of Nominal**
AB03	22-Nov-2024	TKu	HPLC-GLP-9	19, 26, 33, 40	20.7	103.5 %	55.0	91.7 %	4,485.8	99.7 %	9,069.6	100.8 %
				20, 27, 34, 41	21.9	109.5 %	58.1	96.8 %	4,511.6	100.3 %	9,126.2	101.4 %
				21, 28, 35, 42	20.8	104.0 %	58.1	96.8 %	4,510.9	100.2 %	9,062.4	100.7 %
				22, 29, 36, 43	21.8	109.0 %	57.3	95.5 %	4,499.7	100.0 %	9,050.1	100.6 %
				23, 30, 37, 44	19.1	95.5 %	61.0	101.7 %	4,553.0	101.2 %	9,047.6	100.5 %
				24, 31, 38, 45	20.3	101.5 %	57.8	96.3 %	4,518.9	100.4 %	9,045.3	100.5 %
Within-Run Accuracy/Precision III				**Mean**	20.77	103.8 %	57.88	96.5 %	4,513.32	100.3 %	9,066.87	100.7 %
				SD (±)	1.03		1.92		22.62		30.55	
				CV (%)	5.0 %		3.3 %		0.5 %		0.3 %	
				N	6		6		6		6	
Between-Run Accuracy/Precision				**Mean**	20.87	104.3 %	56.20	93.7 %	4,474.68	99.4 %	8,967.08	99.6 %
				SD (±)	1.80		2.33		100.65		237.56	
				CV (%)	8.6 %		4.1 %		2.2 %		2.6 %	
				N	18		18		18		18	

Table 19: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Dog Kidney

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					20.0 ng/g		60.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ±15 % of Nominal**
AB01	14-Nov-2024	LJ	HPLC-GLP-11	105, 112, 119, 126	17.9	89.5 %	56.6	94.3 %	4,370.4	97.1 %	8,616.5	95.7 %
				106, 113, 120, 127	17.9	89.5 %	55.8	93.0 %	4,360.7	96.9 %	8,614.6	95.7 %
				107, 114, 121, 128	17.8	89.0 %	53.3	88.8 %	4,360.6	96.9 %	8,626.5	95.9 %
				108, 115, 122, 129	20.9	104.5 %	53.5	89.2 %	4,394.6	97.7 %	8,618.3	95.8 %
				109, 116, 123, 130	19.2	96.0 %	54.1	90.2 %	4,311.5	95.8 %	8,620.1	95.8 %
				110, 117, 124, 131	18.6	93.0 %	52.6	87.7 %	4,353.1	96.7 %	8,597.8	95.5 %
Within-Run Accuracy/Precision I				**Mean**	18.72	93.6 %	54.32	90.5 %	4,358.48	96.9 %	8,615.63	95.7 %
				SD (±)	1.20		1.56		27.17		9.64	
				CV (%)	6.4 %		2.9 %		0.6 %		0.1 %	
				N	6		6		6		6	
AB02	20-Nov-2024	TKu	HPLC-GLP-9	77, 84, 91, 98	20.8	104.0 %	55.1	91.8 %	4,461.0	99.1 %	9,102.0	101.1 %
				78, 85, 92, 99	22.9	114.5 %	59.2	98.7 %	4,510.1	100.2 %	8,933.6	99.3 %
				79, 86, 93, 100	18.5	92.5 %	54.7	91.2 %	4,487.2	99.7 %	9,099.9	101.1 %
				80, 87, 94, 101	21.6	108.0 %	56.3	93.8 %	4,530.5	100.7 %	8,949.6	99.4 %
				81, 88, 95, 102	19.5	97.5 %	55.6	92.7 %	4,554.8	101.2 %	9,058.2	100.6 %
				82, 89, 96, 103	18.4	92.0 %	57.2	95.3 %	4,531.0	100.7 %	9,145.4	101.6 %
Within-Run Accuracy/Precision II				**Mean**	20.28	101.4 %	56.35	93.9 %	4,512.43	100.3 %	9,048.12	100.5 %
				SD (±)	1.80		1.65		33.92		87.14	
				CV (%)	8.9 %		2.9 %		0.8 %		1.0 %	
				N	6		6		6		6	

Table 19: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Dog Kidney (Continued)

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					20.0 ng/g		60.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ±15 % of Nominal**
AB03	22-Nov-2024	TKu	HPLC-GLP-9	75, 82, 89, 96	20.2	101.0 %	54.0	90.0 %	4,500.9	100.0 %	9,086.6	101.0 %
				76, 83, 90, 97	19.4	97.0 %	59.3	98.8 %	4,541.0	100.9 %	9,103.8	101.2 %
				77, 84, 91, 98	20.1	100.5 %	55.0	91.7 %	4,534.2	100.8 %	8,990.8	99.9 %
				78, 85, 92, 99	22.1	110.5 %	56.8	94.7 %	4,527.8	100.6 %	8,979.9	99.8 %
				79, 86, 93, 100	23.7	118.5 %	57.0	95.0 %	4,539.8	100.9 %	9,013.4	100.1 %
				80, 87, 94, 101	19.5	97.5 %	59.4	99.0 %	4,502.6	100.1 %	9,000.0	100.0 %
Within-Run Accuracy/Precision III				**Mean**	20.83	104.2 %	56.92	94.9 %	4,524.38	100.5 %	9,029.08	100.3 %
				SD (±)	1.71		2.19		18.15		52.66	
				CV (%)	8.2 %		3.9 %		0.4 %		0.6 %	
				N	6		6		6		6	
Between-Run Accuracy/Precision				**Mean**	19.94	99.7 %	55.86	93.1 %	4,465.10	99.2 %	8,897.61	98.9 %
				SD (±)	1.76		2.06		81.83		212.68	
				CV (%)	8.8 %		3.7 %		1.8 %		2.4 %	
				N	18		18		18		18	

Table 20: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Dog Liver

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					20.0 ng/g		60.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ±15 % of Nominal**
AB01	14-Nov-2024	LJ	HPLC-GLP-11	70, 77, 84, 91	18.6	93.0 %	54.7	91.2 %	4,387.9	97.5 %	8,626.6	95.9 %
				71, 78, 85, 92	17.8	89.0 %	56.7	94.5 %	4,369.2	97.1 %	8,679.9	96.4 %
				72, 79, 86, 93	18.4	92.0 %	60.4	100.7 %	4,401.8	97.8 %	8,668.8	96.3 %
				73, 80, 87, 94	17.4	87.0 %	54.6	91.0 %	4,354.2	96.8 %	8,728.3	97.0 %
				74, 81, 88, 95	20.4	102.0 %	56.2	93.7 %	4,353.3	96.7 %	8,683.5	96.5 %
				75, 82, 89, 96	18.8	94.0 %	54.1	90.2 %	4,368.3	97.1 %	8,692.7	96.6 %
Within-Run Accuracy/Precision I				**Mean**	18.57	92.8 %	56.12	93.5 %	4,372.45	97.2 %	8,679.97	96.4 %
				SD (±)	1.04		2.33		19.12		33.12	
				CV (%)	5.6 %		4.1 %		0.4 %		0.4 %	
				N	6		6		6		6	
AB02	20-Nov-2024	TKu	HPLC-GLP-9	49, 56, 63, 70	19.9	99.5 %	56.2	93.7 %	4,548.1	101.1 %	9,174.8	101.9 %
				50, 57, 64, 71	21.0	105.0 %	58.2	97.0 %	4,465.4	99.2 %	9,187.0	102.1 %
				51, 58, 65, 72	18.8	94.0 %	59.1	98.5 %	4,547.6	101.1 %	9,093.0	101.0 %
				52, 59, 66, 73	20.4	102.0 %	56.0	93.3 %	4,568.2	101.5 %	9,125.9	101.4 %
				53, 60, 67, 74	21.9	109.5 %	56.4	94.0 %	4,567.3	101.5 %	9,100.3	101.1 %
				54, 61, 68, 75	22.5	112.5 %	55.8	93.0 %	4,560.1	101.3 %	9,138.0	101.5 %
Within-Run Accuracy/Precision II				**Mean**	20.75	103.8 %	56.95	94.9 %	4,542.78	101.0 %	9,136.50	101.5 %
				SD (±)	1.35		1.36		38.95		38.30	
				CV (%)	6.5 %		2.4 %		0.9 %		0.4 %	
				N	6		6		6		6	

Table 20: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Dog Live (Continued)

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					20.0 ng/g		60.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ±15 % of Nominal**
AB03	22-Nov-2024	TKu	HPLC-GLP-9	47, 54, 61, 68	20.3	101.5 %	57.5	95.8 %	4,488.4	99.7 %	8,976.2	99.7 %
				48, 55, 62, 69	23.0	115.0 %	58.6	97.7 %	4,508.8	100.2 %	8,994.3	99.9 %
				49, 56, 63, 70	21.6	108.0 %	56.2	93.7 %	4,522.4	100.5 %	8,972.9	99.7 %
				50, 57, 64, 71	21.3	106.5 %	58.5	97.5 %	4,509.5	100.2 %	8,863.9	98.5 %
				51, 58, 65, 72	21.8	109.0 %	56.8	94.7 %	4,499.9	100.0 %	8,973.0	99.7 %
				52, 59, 66, 73	20.4	102.0 %	56.5	94.2 %	4,433.6	98.5 %	9,026.1	100.3 %
Within-Run Accuracy/Precision III				**Mean**	21.40	107.0 %	57.35	95.6 %	4,493.77	99.9 %	8,967.73	99.6 %
				SD (±)	1.00		1.03		31.56		54.81	
				CV (%)	4.7 %		1.8 %		0.7 %		0.6 %	
				N	6		6		6		6	
Between-Run Accuracy/Precision				**Mean**	20.24	101.2 %	56.81	94.7 %	4,469.67	99.3 %	8,928.07	99.2 %
				SD (±)	1.64		1.65		79.21		198.12	
				CV (%)	8.1 %		2.9 %		1.8 %		2.2 %	
				N	18		18		18		18	

Table 21: Dilution Integrity for the Analysis of 2439-exNA in Dog Brain

		QC-500xULOQ (Dilution 1:1,000)	
		5,000,000.0 ng/g	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/g**	**Acc. Criterion ± 15 % of Nominal**
AB01	133	3,573,430.3	71.5 %
	134	4,718,549.2	94.4 %
	135	4,695,085.2	93.9 %
	136	4,708,105.8	94.2 %
	137	4,416,959.5	88.3 %
	138	3,738,497.1	74.8 %
	Mean	4,308,437.85	86.2 %
	SD (±)	520,435.87	
	CV (%)	12.1 %	
	N	6	

Underlined QC sample did not fulfill the acceptance criterion but was included in the statistical calculations (Mean, SD and CV).

Table 22: Long-Term Stability for the Analysis of 2439-exNA in Dog Brain Homogenate

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
			Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal
AB01	T= 0 (-80 °C ± 20 °C)	14, 17	56.5	94.2 %	9,076.2	100.8 %
		15, 18	54.7	91.2 %	8,949.1	99.4 %
		16, 19	56.3	93.8 %	9,021.3	100.2 %
		Mean	55.83	93.1 %	9,015.53	100.2 %
		SD (±)	0.99		63.75	
		CV (%)	1.8 %		0.7 %	
		N	3		3	

Table 23: Long-Term Stability for the Analysis of 2439-exNA in Dog Brain Lysate

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
			Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal
AB01	T= 0 (-80 °C ± 20 °C)	21, 24	59.0	98.3 %	9,035.8	100.4 %
		22, 25	56.9	94.8 %	8,983.8	99.8 %
		23, 26	59.3	98.8 %	9,003.1	100.0 %
		Mean	58.40	97.3 %	9,007.57	100.1 %
		SD (±)	1.31		26.29	
		CV (%)	2.2 %		0.3 %	
		N	3		3	

Table 24: Freeze-Thaw Stability for the Analysis of 2439-exNA in Dog Brain Lysate

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
			Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal
AB03	• FTS: 5 cycles (-80 °C ± 20 °C)	103, 106	58.3	97.2 %	9,020.4	100.2 %
		104, 107	64.1	106.8 %	9,059.1	100.7 %
		105, 108	61.6	102.7 %	9,257.7	102.9 %
		Mean	61.33	102.2 %	9,112.40	101.2 %
		SD (±)	2.91		127.31	
		CV (%)	4.7 %		1.4 %	
		N	3		3	

Table 25: Bench-Top Stability/Robustness and Short-Term Stability for the Analysis of 2439-exNA in Dog Brain Lysate

Run ID	Validation Parameters	Sample No.	QC-Low		QC-High	
			60.0 ng/g		9,000.0 ng/g	
			Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal
AB02	2 hours at room temperature after Proteinase K digestion and 2 hours at room temperature after SDS precipitation	111, 114	57.0	95.0 %	8,656.8	96.2 %
		112, 115	55.0	91.7 %	8,764.0	97.4 %
		113, 116	54.6	91.0 %	8,905.4	98.9 %
		Mean	55.53	92.6 %	8,775.40	97.5 %
		SD (±)	1.29		124.69	
		CV (%)	2.3 %		1.4 %	
		N	3		3	

Table 26: Processed Sample Stability for the Analysis of 2439-exNA in Dog Brain

Run ID	Date of Analysis	Sample No.	QC-Low 60.0 ng/g		QC-High 9,000.0 ng/g	
			Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal
AB02	20-Nov-2024 (Determination 1)	14, 17	56.8	94.7 %	9,267.4	103.0%
		15, 18	56.3	93.8 %	9,281.1	103.1%
		16, 19	54.8	91.3 %	9,322.8	103.6%
		Mean	55.97	93.3 %	9,290.43	103.2%
		SD (±)	1.04		28.86	
		CV (%)	1.9 %		0.3 %	
		N	3		3	
AB02	22-Nov-2024 (Determination 2, covering 37 hours)	118, 121	54.7	91.2 %	9,245.4	102.7%
		119, 122	53.5	89.2 %	9,211.1	102.3%
		120, 123	55.9	93.2 %	9,239.0	102.7%
		Mean	54.70	91.2 %	9,231.83	102.6%
		SD (±)	1.20		18.24	
		CV (%)	2.2 %		0.2 %	
		N	3		3	

Table 26: Processed Sample Stability for the Analysis of 2439-exNA in Dog Brain (Continued)

Run ID	Date of Analysis	Sample No.	QC-Low 60.0 ng/g		QC-High 9,000.0 ng/g	
			Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal	Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 15 % of Nominal
AB03	22-Nov-2024 (Determination 3, covering 46 hours)	12, 15	58.9	98.2 %	9,367.8	104.1 %
		13, 16	57.2	95.3 %	9,389.1	104.3 %
		14, 17	58.1	96.8 %	9,453.9	105.0 %
		Mean	58.07	96.8 %	9,403.60	104.5 %
		SD (±)	0.85		44.84	
		CV (%)	1.5 %		0.5 %	
		N	3		3	
AB03	25-Nov-2024 (Determination 4, covering 119 hours)	118, 121	56.2	93.7 %	9,273.2	103.0 %
		119, 122	56.5	94.2 %	9,267.7	103.0 %
		120, 123	56.2	93.7 %	9,312.0	103.5 %
		Mean	56.30	93.8 %	9,284.30	103.2 %
		SD (±)	0.17		24.15	
		CV (%)	0.3 %		0.3 %	
		N	3		3	

Table 27: Short-Term Stability of Working Solution for the Analysis of 2439-exNA in Dog Tissue

Run ID	Validation Parameters	Sample No.	Working Solution LLOQ 20.0 ng/g	
			Calculated 2439-exNA Conc. ng/g	Acc. Criterion ± 20 % of Nominal
AB01	4 hours at room temperature	140	18.9	94.5 %
		141	18.6	93.0 %
		142	19.0	95.0 %
		Mean	18.83	94.2 %
		SD (±)	0.21	
		CV (%)	1.1 %	
		N	3	

14 SUPPLEMENTARY INFORMATION

Supplemental File 1: Axolabs *MDR_2439exNA_DogTissue* (edition 01, effective date: 24-Oct-2024)

Supplemental File 2: Axolabs *STM_Broad_2439-exNA_DogTissue* (edition 02, effective date: 03-Dec-2024)